

NEURAL BELLMAN OPERATORS

QIAN QI

Peking University

We study Bellman solution procedures that separate neural proposals, economically meaningful improvement, and independent certification. A backward residual-budget operator preserves candidate actions whenever the completed continuation leaves sufficient accuracy allowance. We prove all-state, all-restart accuracy and pointwise payoff improvement without an optimal-value reference. Exact-rational experiments audit 115 frozen proposals and 286 completions, including trivial-policy controls and graphs with up to 16,384 states. For continuous control, precision-aware coordinate polling certifies projected stationarity at active box constraints. A likelihood-ratio oracle instantiates its assumptions in a stopped consumption–portfolio economy with costly preference adjustment. Independently implemented historical transport and paired rollback experiments identify the roles of parameter geometry and deployment checks. Classical comparators, raw failures, and verification costs remain explicit. The original whole-domain accuracy target is retained separately from the proved finite-state and restricted-class results.

KEYWORDS: Bellman operators, Stochastic control, Certified computation, Preference adjustment, Neural approximation.

1. INTRODUCTION

Numerical dynamic programming must answer two different questions. Can an approximator discover a useful policy, and can the computation establish that the policy is economically accurate? A small training loss, a favorable simulated return, and a smooth optimizer trajectory do not answer the second question when stopping, financing restrictions, and active controls are present. Conversely, a rigorous certificate for a known policy does not demonstrate that an algorithm can discover the solution of an unknown Bellman problem. This paper joins discovery and certification through explicitly specified operators, rather than identifying either component with the other.

Our organizing economic problem is a consumption–portfolio economy in which the agent can alter a stochastic preference state at a quadratic cost. Consumption, investment, preference adjustment, and the continuation region interact. We retain the original objective: a current-state policy whose regret is at most .01 uniformly over the economic state domain and all restart times. The revision adds three constructive results without changing that objective. First, a backward Bellman completion operator turns a learned finite-state candidate into a policy with a prescribed uniform regret bound. The operator uses its own policy values and local action comparisons, not an optimal-value oracle. Second, a precision-aware direct-search result provides projected-gradient stationarity at box constraints. An explicit likelihood-ratio bridge makes its smoothness and oracle requirements numerically checkable on a slice of the original stopped economy. Third, an exact recentering identity isolates finite-path neural Jacobian and optimizer-moment transport, including the nonlinear output displacement omitted by a linearized control.

The computations are designed to test these statements separately. A stopped, coupled inventory problem supplies unknown-solution Bellman discovery and exact policy evaluation. A constant-control slice of the original economy supplies active-boundary and interior economic optima. Matched production-schedule experiments supply actual deployment decisions, including optimizer-state restoration. The inherited stochastic manufactured example remains a

verification stress test, and the inherited full-domain experiment remains the test of the original objective. We do not substitute the former for the latter.

This distinction is quantitatively consequential. All 34 inventory candidates meet .01 after the same completion rule, whereas only 12 meet it before completion. Yet exact tabular backward induction is also available at these sizes and is charged as a direct baseline. The matched central-state frontier does not identify neural superiority. The best retained whole-domain bound for the original economy is 7.181834580823298, approximately 718.18 times its stated tolerance. Our new accuracy results concern precisely the problems and policy classes for which their hypotheses are instantiated. They are neither a numerical proof of that original target nor a statistical claim about optimizer superiority.

1.1. *Relation to the literature and the incremental contribution*

Classical computational economics emphasizes approximation design, economic feasibility, and error control; see Judd (1998). Markov-chain approximation for continuous stochastic control provides a systematic discretization theory under its own consistency and convergence conditions (Kushner and Dupuis, 2001). Adaptive sparse grids offer a major classical response to high-dimensional economic state spaces (Brumm and Scheidegger, 2017). Our finite inventory graph is an independently specified economic model, not a consistency-verified discretization of the original diffusion. Its exact finite-state accuracy therefore makes no assertion about a continuous-domain discretization error or a sparse-grid performance comparison.

Fitted value iteration propagates approximation error through Bellman recursion; finite-sample bounds quantify that propagation under distributional and function-class conditions (Munos and Szepesvári, 2008). Deep stochastic-control algorithms study approximation and generalization in finite-horizon problems (Huré, Pham, Bachouch, and Langrené, 2021). Neural PDE methods provide mesh-free residual approximation (Sirignano and Spiliopoulos, 2018). These works motivate the discovery stage, but neither a sampled residual nor a convergence statement with an optimization-error term is the same object as a fixed-candidate, all-state arithmetic certificate. Our inventory implementation performs fitted Bellman recursion without a supplied value witness, freezes its actions, and then constructs a deterministic residual envelope and a reference-free corrective sweep. Bellman residual bounds and policy improvement are classical principles. The contribution here is their explicit tolerance-preserving composition, exact arithmetic implementation, and separation from both learned policy generation and a subsequent independent optimal reference.

Information-relaxation duality bounds stochastic dynamic programs by relaxing nonanticipativity and charging penalties (Brown, Smith, and Sun, 2010, Brown and Smith, 2014). Our stopped supersolution argument is complementary: it bounds a policy and an optimal value using a smooth witness, a boundary mismatch, and generator residuals. Neither form of upper bound makes a candidate feasible automatically. The financed policy construction is evaluated only after admissibility is checked in the original information filtration. The historical dual library is a shared input with a separately identified construction cost, not a free neural training oracle.

Constrained derivative-free optimization is well developed. Mesh adaptive direct search treats general constraints and nonsmooth stationarity under appropriate assumptions (Audet and Dennis, 2006); direct-search sufficient-decrease arguments give worst-case iteration bounds (Vicente, 2010). We do not claim to introduce constrained direct search. Our box result exposes the cost of an interval oracle at the necessary precision and gives a short-face projected-gradient bound: geometrically unavailable full steps are treated through projection, while computationally unresolved evaluations cannot establish stationarity. The resulting theorem is instantiated below with validated derivatives and payoff enclosures, rather than inferred from a software test.

Adam uses a history of coordinatewise moments (Kingma and Ba, 2015). Limited-memory quasi-Newton methods are serious classical controls, not optional weak baselines (Byrd, Lu, Nocedal, and Zhu, 1995). Function-space kernel descriptions of neural learning are central to the neural tangent kernel literature (Jacot, Gabriel, and Hongler, 2018); our identity instead concerns a finite-width network, a realized finite step, and the complete historical Adam state. It is an exact chain-rule comparison, not a fixed-kernel or infinite-width limit. The independent recentered implementation retains both the network carrier and the nonlinear output remainder, so its agreement with neural Adam identifies a mechanism without establishing a cheaper nonneural solver.

Finally, rigorous floating-point verification requires explicit control of rounding, truncation, and domain coverage (Rump, 2010). We distinguish three arithmetic objects: exact rational finite-state policies; mathematical real networks with saved binary64 coefficients interpreted exactly; and hardware floating-point deployment. Only the first two are certified here. A hardware execution theorem would require an additional rounding enclosure at deployment.

2. THE ECONOMIC PROBLEM AND THE CERTIFICATE TARGET

Time runs from 0 to $T = 1$. Before stopping, preference u and wealth x evolve according to

$$du_s = \theta_s ds + .05 dZ_s^u, \quad (1)$$

$$dx_s = \{(.02 + .06p_s)x_s - c_s\} ds + .2p_s x_s dZ_s^x, \quad (2)$$

where $d\langle Z^u, Z^x \rangle_s = -.25 ds$. Controls satisfy

$$c \in [.05, .8], \quad \theta \in [-.2, .2], \quad p \in [-.5, .8]. \quad (3)$$

The continuation domain is $\mathcal{D} = (1.2, 2.8) \times (.5, 2)$. Let τ be first exit from $\mathcal{D}$, capped at 1. The running payoff and settlement are

$$\ell_k(u, c, \theta) = \frac{c^{1-u}}{1-u} - \frac{k}{2} \theta^2, \quad k \in [.5, 8], \quad (4)$$

$$G(u, x) = -.02(u-2)^2 + .1 \log x, \quad g(s, u, x) = G(u, x) - 8(1-s). \quad (5)$$

At $s = 1$, $g = G$. An admissible progressively measurable policy, adapted to the original Brownian filtration, has payoff

$$J_k^\pi(t, u, x) = \mathbb{E}_{t, u, x} \left[\int_t^\tau e^{-.04(s-t)} \ell_k(u_s, c_s, \theta_s) ds + e^{-.04(\tau-t)} g(\tau, u_\tau, x_\tau) \right]. \quad (6)$$

Write $V_k = \sup_{\pi \in \mathcal{A}} J_k^\pi$. The unrestricted value includes admissible path-dependent policies; the target implementation is a single current-state actor $\pi(t, u, x)$. The original full-domain numerical experiment fixes $k = 2$ and asks for

$$\sup_{(t, u, x) \in [0, 1] \times \overline{\mathcal{D}}} \{V_2(t, u, x) - J_2^\pi(t, u, x)\} \leq .01. \quad (7)$$

The boundary values use immediate settlement. We retain (7) throughout. A certificate at $(0, 2, 1.25)$, a certificate on a small restart rectangle, or an optimum within a constant-control class is not this statement.

The adjustment cost has a direct economic role. Positive θ changes future utility curvature, but its flow cost rises with k . Stopping adds a substantial loss before the horizon. Consequently

a policy can have accurate typical-state behavior and still have large worst-restart regret near financially or psychologically fragile states. This is the reason to report a uniform criterion, rather than a mean simulation score alone.

2.1. Stopped residual verification

For a smooth witness v , define

$$\begin{aligned} \mathcal{R}^a v = & v_t + \theta v_u + \{(.02 + .06p)x - c\}v_x + .00125v_{uu} + .02p^2x^2v_{xx} - .0025pxv_{ux} \\ & + \ell_k(u, c, \theta) - .04v. \end{aligned} \quad (8)$$

No differentiability of V_k is required in the next statement.

PROPOSITION 1—Uniform policy certificate: *Suppose $v \in C^{1,2}$ has the boundedness needed for stopped Itô’s formula on the closed cylinder. Suppose the policy is admissible, $|v - g| \leq b$ on the parabolic boundary, and, throughout the continuation cylinder,*

$$|\mathcal{R}^\pi v| \leq e, \quad \sup_{a \in \mathcal{A}} \mathcal{R}^a v - \mathcal{R}^\pi v \leq q, \quad (9)$$

where $b, e, q \geq 0$. Then

$$0 \leq V_k(t, u, x) - J_k^\pi(t, u, x) \leq 2b + (2e + q) \frac{1 - e^{-.041-t}}{.04}. \quad (10)$$

State-dependent and slabwise versions improve this bound when signed residual information is available. The all-restart suffix envelope used in the retained full-state calculation is proved in Appendix E. A numerical grid of test points alone cannot establish any of these hypotheses; the intervals must cover each whole cell and each feasible control range.

2.2. Feasible memory and the financed comparison

The retained financed family uses sixteen time slabs. Its network has 355 parameters and 48 raw outputs. A common raw utility intercept is absorbed by the financing root, leaving 47 economic quotient coordinates. This quotient removes a redundancy; it does not turn every neural parameter displacement into an independent economic direction.

The construction can retain the adapted internal state

$$dY_s = .02Y_s ds - .3Y_s dZ_s^x, \quad (11)$$

with expert wealth processes $X_i = F_i(s, Y_s)$ and portfolio fractions $p_i = -1.5Y_s F_{i,y}/F_i$. Non-negative initial mixture weights yield aggregate wealth and wealth-weighted investment, subject to the inherited feasibility checks. This policy is Markov in an augmented state containing Y and the fixed mixture weights, not necessarily in (t, u, x) alone. Since the augmentation is generated from the original observations and carries no independent randomness, it does not enlarge the original admissible filtration. The unrestricted augmented and original control problems can therefore represent the same admissible policies, while the particular finite financed library remains a restricted class.

A central financed certificate is useful evidence about proposal generation and deployment. It cannot establish (7) for a current-state actor. We preserve that distinction in the production tables, the historical comparison, and the response to the referee.

3. CERTIFIED BELLMAN AND DEPLOYMENT OPERATORS

We distinguish a proposal operator, a completion operator, and a deployment operator. The first may be neural and approximate. The second uses a Bellman error criterion on a specified state model. The third compares economic payoffs and restores the entire optimizer state after rejection. Their conclusions are different: approximation quality, uniform policy regret, and monotone deployed payoff, respectively.

3.1. Reference-free Bellman completion

Consider a finite-horizon Markov decision problem on finite state sets S_t , with nonempty feasible actions $A_t(s)$, discount $\beta \in (0, 1]$, known transition probabilities, bounded rewards, and a fixed terminal payoff. A stopped state has only its settlement action and zero continuation. Let

$$(T_tv)(s) = \max_{a \in A_t(s)} \left\{ r_t(s, a) + \beta \sum_{s'} P_t(s' | s, a) v(s') \right\}. \quad (12)$$

For a deterministic policy π , compute its own value exactly by backward evaluation, $v_t^\pi = T_t^\pi v_{t+1}^\pi$. Neither this evaluation nor a comparison of its local actions requires V^* . Define the nonnegative Bellman deficit

$$\delta_t^\pi(s) = (T_tv_{t+1}^\pi)(s) - v_t^\pi(s), \quad (13)$$

and its state-sensitive envelope

$$B_T^\pi = 0, \quad B_t^\pi(s) = \delta_t^\pi(s) + \beta \max_{a \in A_t(s)} \sum_{s'} P_t(s' | s, a) B_{t+1}^\pi(s'). \quad (14)$$

Both quantities are zero at a stopped state.

THEOREM 1—Residual envelope and uniform completion: *For every deterministic policy, $0 \leq V_t^*(s) - v_t^\pi(s) \leq B_t^\pi(s)$ at every state and restart time. Fix $\varepsilon > 0$ and*

$$\eta = \varepsilon \left(\sum_{j=0}^{T-1} \beta^j \right)^{-1}. \quad (15)$$

Given any raw policy π^0 , construct a completed policy backward. At time t , evaluate each action against the already completed continuation $v_{t+1}^{\pi^0}$. Retain $\pi_t^0(s)$ if its deficit from the largest such action value is at most η ; otherwise select a maximizing feasible action. Then

$$0 \leq V_t^*(s) - v_t^{\bar{\pi}}(s) \leq \eta \sum_{j=0}^{T-t-1} \beta^j \leq \varepsilon \quad (16)$$

for all states and restarts. The algorithm uses no optimal-value reference. With at most N states, A actions, and D successor outcomes, each evaluation, envelope, and completion sweep uses $O(TNAD)$ arithmetic operations. Exact-integer bit complexity additionally depends on the bit lengths of rewards, probabilities, discount factors, and accumulated values.

The proof is a backward comparison argument, given in Appendix A. The terminal stopping convention is essential: a settled state cannot be improved by continuing the chain. In the implementation, exact rational deficits, rather than their printed decimal representations, determine both action replacement and the tolerance test.

The completed policy is the compiled action table, with the changed states recorded. It is not generally the raw neural network evaluated in floating-point arithmetic. In particular, the theorem must not be described as a certificate for an unchanged neural policy when completion replaces an action. The neural representation supplies the candidate value recursion, and completion preserves its decision wherever the suffix-conditioned Bellman deficit is sufficiently small. This construction permits an exact accuracy guarantee without assuming that neural optimization finds a global minimum.

The operation count also reveals the principal computational qualification. At the state sizes studied here, exact backward induction has the same enumeration order. Completion is not a new way to evade tabular complexity. The experiment tests whether discovery and certification can be combined without reference leakage; the classical reference and its cost remain visible.

3.2. Synchronous economic deployment

Let S contain the policy parameters and all optimizer state, including moment estimates, step counters, and the rule’s current hyperparameters. Suppose a verifier returns $[L(\pi), U(\pi)]$ containing the exact economic payoff of a feasible policy at the declared restart. A proposal generated from S is deployed only if

$$L(\pi_{\text{candidate}}) > U(\pi_{\text{incumbent}}) + \zeta, \quad \zeta \geq 0. \quad (17)$$

Otherwise the saved S is restored. A verifier failure gives no usable interval and is not an acceptance.

PROPOSITION 2—Monotone deployment: *Every accepted proposal improves exact payoff by more than ζ . Rejection followed by exact state restoration leaves both the deployed policy and the next proposal’s initial optimizer state equal to their saved values. If $\zeta > 0$ and payoffs are bounded above by M , at most $\lfloor (M - J_0)/\zeta \rfloor$ accepted proposals can occur.*

For $\zeta = 0$, monotonicity does not imply a finite number of acceptances, convergence of Adam, or discovery of a global optimum. Nor is (17) equivalent to comparing two regret upper bounds. A tighter upper bound can result from a better witness or finer verifier while the candidate’s actual payoff falls.

An interval pair can overlap even when the candidate is genuinely better. If a true improvement is $\gamma > 0$ and each interval has width at most w , separation is guaranteed when $2w < \gamma$. Refining widths $w_j = w_0 2^{-j}$ therefore resolves a fixed positive margin after finitely many successful verifier levels. Zero margin, a finite precision floor, or an exhausted resource budget leaves the comparison unresolved; it is not evidence of economic indifference.

3.3. Box-constrained stationarity with explicit oracle work

Let $C = \prod_{i=1}^n [\ell_i, u_i]$ and let $F : C \rightarrow \mathbb{R}$ be differentiable with L -Lipschitz gradient on the relevant feasible segments. For $\tau > 0$, define the ascent projected-gradient mapping

$$G_\tau(x) = \frac{\Pi_C(x + \tau \nabla F(x)) - x}{\tau}. \quad (18)$$

Its zero set is the first-order box-stationary set. The gradient need not be zero at a constrained optimum.

At mesh $h > 0$, a coordinate poll evaluates every feasible full step $x \pm he_i$. The interval oracle must enclose F with width at most κh^2 at the incumbent and at every feasible trial used in a complete unsuccessful poll. A step is accepted if its lower endpoint exceeds the incumbent's upper endpoint by σh^2 , where $\sigma > 0$. A full step outside C is recorded as a geometric short face with its distance to the boundary. A feasible step whose verification is unresolved is not a short face.

THEOREM 2—Constrained precision-aware polling: *Suppose $F \leq M$ on the feasible search region and the stated width is obtained whenever required, at work at most $W(\kappa h^2)$ per call. Starting from x_0 , repeated accepted coordinate steps followed by complete polls reach an unsuccessful complete poll using work at most*

$$(2n + 1) \left[1 + \left\lfloor \frac{M - F(x_0)}{\sigma h^2} \right\rfloor \right] W(\kappa h^2). \quad (19)$$

At its final point x ,

$$\|G_\tau(x)\|_2 \leq \sqrt{n} h \max \left\{ \sigma + 2\kappa + \frac{L}{2}, \frac{1}{\tau} \right\}. \quad (20)$$

If any required feasible evaluation fails to reach the requested width, the conclusion is oracle unresolved, not (20).

The maximum in (20) is the boundary term missing from an interior positive-spanning argument. When a full step exists in the sign of a gradient component, unsuccessful comparison bounds that component by $(\sigma + 2\kappa + L/2)h$. When it does not, the distance to the corresponding face is less than h , so projection bounds the component of G_τ by h/τ . This proves constrained stationarity without treating an infeasible direction as a failed economic comparison.

Equation (19), not a bare $O(\varepsilon^{-2})$ poll count, is the complexity statement. Choosing h proportional to a desired projected-gradient tolerance gives an iteration factor of order ε^{-2} but requires width of order ε^2 . A rapidly growing W can dominate all proposal work. The theorem does not establish arbitrary precision for the original 47-dimensional stopped checker. Our implementation below verifies the needed widths at four declared meshes on an original-economy slice, and reports actual setup, derivative-audit, and oracle work. It does not extrapolate a fixed truncation order to arbitrarily small meshes.

3.4. Exact recentered historical transport

Let $f(\theta) \in \mathbb{R}^{48}$ denote the raw network outputs, and let $Q \in \mathbb{R}^{48 \times 47}$ have orthonormal columns spanning the economically relevant quotient. Locally the proposal loss depends on raw outputs only through the quotient after the financing intercept has been eliminated. Write g_r for its quotient gradient and $B_r = Q^\top Df(\theta_r)$. For minimization, Adam's first moment is

$$\hat{m}_n = \sum_{r=1}^n a_{nr} B_r^\top g_r, \quad a_{nr} = \frac{(1 - \beta_1) \beta_1^{n-r}}{1 - \beta_1^n}, \quad (21)$$

with the corresponding coordinatewise second moment and denominator D_n . Thus

$$f(\theta_{n+1}) - f(\theta_n) = -\alpha \sum_{r=1}^n a_{nr} Df(\theta_n) D_n^{-1} B_r^\top g_r + R_n, \quad (22)$$

where R_n is the exact nonlinear Taylor remainder. If Df is H -Lipschitz on the step segment, $\|R_n\| \leq H\|\theta_{n+1} - \theta_n\|^2/2$. The exact identity does not require this bound to be small.

PROPOSITION 3—Recentered path equivalence: *Initialize two implementations with identical carrier parameters, Adam moments, counters, and hyperparameters. In the second implementation evaluate the economic gradient at the carrier’s realized raw output, pull it back using $B_n^\top$, apply the same parameter-space Adam recurrence, and recenter the next economic coordinates on the full realized output $f(\theta_{n+1})$. In exact arithmetic the two implementations have identical parameters, moments, raw outputs, and proposal policies at every step, including the entire nonlinear remainder in (22).*

This proposition is an induction using the chain rule. It does not say that a 47-dimensional optimizer with no neural carrier reproduces 355-dimensional Adam, nor does it imply a chart-independent adaptive optimizer. The implementation uses two independently coded recurrences and measures their floating-point discrepancy. Exact real equality is a mathematical result; the measured discrepancy is a consistency check, not a substitute for its assumptions.

The previous moving control used a capped linearized output update, omitted R_n , and let the economic quotient drift from the carrier. Its poor performance consequently confounded transport, clipping, nonlinear displacement, and misalignment. The new control removes those confounds by construction. Agreement identifies what the neural trajectory is doing; it cannot by itself establish that the neural representation is necessary or cheaper.

4. STATE-DEPENDENT RESIDUAL BUDGETS

The uniform local tolerance in Theorem 1 reserves the same amount for every possible future stage. Once a continuation has been completed, however, its actual residual envelope is known. We can allocate only the amount needed by that continuation, and test the raw action against the remaining budget. This distinction concerns preservation of candidate decisions, not a change in the information available to the algorithm.

Let $e_t : S_t \rightarrow [0, \infty)$ be a prescribed family of error budgets, with $e_T = 0$ and zero budget at settled states. Assume that, at every nonsettled state,

$$e_t(s) \geq \beta \max_{a \in A_t(s)} \sum_{s'} P_t(s' | s, a) e_{t+1}(s'). \quad (23)$$

A common tolerance $e_t(s) = \varepsilon$ before settlement satisfies this condition when $\beta \leq 1$. During the backward construction, let v_{t+1} and B_{t+1} be the exact value and residual envelope of the already completed suffix. Set

$$q_t(s, a) = r_t(s, a) + \beta P_{t,s,a} v_{t+1}, \quad (24)$$

$$c_t(s) = \beta \max_{a \in A_t(s)} P_{t,s,a} B_{t+1}, \quad a_t^{\text{rem}}(s) = e_t(s) - c_t(s). \quad (25)$$

Retain the raw action $a^0 = \pi_t^0(s)$ exactly when

$$\max_a q_t(s, a) - q_t(s, a^0) \leq a_t^{\text{rem}}(s). \quad (26)$$

Otherwise choose a maximizing feasible action, using a fixed tie rule. For the selected action a , set $v_t(s) = q_t(s, a)$ and

$$B_t(s) = \max_b q_t(s, b) - q_t(s, a) + c_t(s). \quad (27)$$

At a settled state, apply its settlement payoff and set $B_t = 0$; the terminal value is the stated terminal payoff and $B_T = 0$.

THEOREM 3—Budgeted completion and policy preservation: *Under the finite-state assumptions of Theorem 1 and (23), the backward construction is well defined. At every state and restart,*

$$v_t^{\pi^0}(s) \leq v_t^{\bar{\pi}}(s) \leq V_t^*(s), \quad 0 \leq V_t^*(s) - v_t^{\bar{\pi}}(s) \leq B_t(s) \leq e_t(s). \quad (28)$$

The local remaining allowance is nonnegative. Conditional on the already completed suffix and on using envelope (27), the rule retains the raw action whenever that envelope can certify it within $e_t(s)$. No optimal value or optimal action reference is needed. Its arithmetic enumeration order is $O(TNAD)$; exact bit complexity and storage are charged separately.

PROOF: The assertion holds at the terminal time and at settlement. Suppose it holds at $t + 1$. Since $0 \leq B_{t+1} \leq e_{t+1}$ and transition probabilities are nonnegative, (23) gives $0 \leq c_t(s) \leq e_t(s)$. Thus the allowance in (25) is nonnegative. If the raw action passes (26), adding c_t proves $B_t \leq e_t$. If it fails, a maximizing feasible action exists by finiteness; its local deficit is zero, so $B_t = c_t \leq e_t$.

For the accuracy statement, write $d_{t+1} = V_{t+1}^* - v_{t+1}^{\bar{\pi}}$. By the induction hypothesis $0 \leq d_{t+1} \leq B_{t+1}$. The Bellman equation and the elementary inequality $\max_a (x_a + y_a) \leq \max_a x_a + \max_a y_a$ yield

$$V_t^* - v_t^{\bar{\pi}} = \max_a \{q_t(s, a) + \beta P_{t,s,a} d_{t+1}\} - q_t(s, \bar{\pi}_t(s)) \leq B_t(s).$$

The loss is nonnegative by optimality of V^* . For preservation, assume $v_{t+1}^{\bar{\pi}} \geq v_{t+1}^{\pi^0}$. The selected action has $q_t(s, \bar{\pi}_t(s)) \geq q_t(s, \pi_t^0(s))$, because it is either the raw action or a maximizer. Positivity of the transition law then gives

$$v_t^{\bar{\pi}}(s) \geq r_t(s, \pi_t^0(s)) + \beta P_{t,s,\pi_t^0(s)} v_{t+1}^{\pi^0} = v_t^{\pi^0}(s).$$

This closes the induction. Finally, once v_{t+1} and B_{t+1} are fixed, c_t does not depend on the selected current action. The raw action satisfies $B_t \leq e_t$ if and only if (26) holds. Computing the action values and propagating the envelope each enumerates at most $TNAD$ successor contributions. None of these steps calls an optimal-reference solver. *Q.E.D.*

The preservation inequality also holds for the uniform-threshold operator, by the same induction. It is a pointwise economic comparison on the finite model, stronger than observing that a printed regret bound decreased. It does not carry over automatically to a continuous-state economy or to a floating-point neural implementation.

The theorem is locally permissive, not globally minimum-edit. Changing a later action changes both its value and its envelope, which can change earlier decisions. Consequently we make no claim that the budget rule changes fewer actions on every input, nor that either rule yields the cheapest certified policy. The two rules are compared on exactly the same raw tables. A fixed common tolerance has the advantage of certifying every restart, rather than just one initial distribution. More general budgets satisfying (23) can incorporate state-specific accuracy priorities.

This construction uses the full model, feasible-action enumeration, and exact suffix evaluation. These are the same information resources available to exact backward induction. Classical dynamic programming supplies the Bellman comparison argument; the methodological addition

here is the explicit candidate-preservation rule, its state-dependent certificate budget, and the joint proof of local permissiveness, whole-state accuracy, and payoff monotonicity. It is not an invention of policy iteration or an unconditional neural acceleration theorem.

4.1. Relation to established numerical methods

The oracle and conclusion must be matched before comparing methods. Fitted value iteration propagates approximation and sampling errors through a value recursion; Munos and Szepesvári (2008) give finite-sample bounds under their approximation and distributional assumptions. Our finite graph instead permits exact model evaluation after a policy is frozen. Its deterministic, all-state certificate is obtained by enumeration; it cannot be advertised as a better statistical rate or as a solution of the dimensionality problem addressed by approximation methods.

Constrained direct search predates the present work. Kolda, Lewis, and Torczon (2003) organize direct-search methods and their convergence mechanisms; Kolda, Lewis, and Torczon (2006) relate a stationarity measure to the step-length control in generating-set search under linear constraints. The contribution of Theorem 2 is the explicit interval-width requirement, the short-face alternative for a box, the work factor $W(\kappa h^2)$, and the unresolved-oracle outcome. The theorem is an instantiated certificate argument, not a claim to have originated constrained derivative-free optimization.

The verified-numeric literature, represented by Rump (2010), supplies the reason for using directed enclosures rather than tolerances on floating-point residuals. The stopped slice contributes a concrete analytic oracle: change of path measure controls the stopping discrepancy, while polynomial moments, Gaussian tails, and directed rounding enclose the objective and its first two derivatives. Neither neural stochastic-control approximation (Huré, Pham, Bachouch, and Langrené, 2021) nor a small training residual alone supplies those particular deterministic enclosures. Conversely, the slice has one constant preference-control parameter and does not establish a general high-dimensional oracle.

The comparative economic benchmark remains the unchanged consumption–portfolio model. Dual bounds and information relaxation (Brown, Smith, and Sun, 2010, Brown and Smith, 2014), classical constrained quasi-Newton optimization (Byrd, Lu, Nocedal, and Zhu, 1995), and projection and sparse-grid methods (Judd, 1998, Brumm and Scheidegger, 2017) are substantive comparators, not incidental implementation choices. The numerical contribution is identified by theorem, oracle, coupling rule, and executed application separately. A general claim of neural superiority would require a favorable accuracy–total-work comparison; the present revision reports that comparison rather than assuming its outcome.

5. AN INSTANTIATED CERTIFICATE IN THE ORIGINAL STOPPED ECONOMY

Theorem 2 becomes a computational result only after the payoff oracle, smoothness constant, feasible geometry, and precision are supplied. We do so without changing the primitives in Section 2. Fix the restart $(0, 2, 5/4)$, consumption $c = 3/4$, investment $p = 0$, and constant adjustment $\theta \in [-1/5, 1/5]$. We examine $k = 1/2$ and $k = 2$. This is an admissible one-dimensional class inside the original economy, not a replacement for all current-state policies.

Wealth is deterministic:

$$x_s = \left(\frac{5}{4} - \frac{3/4}{.02} \right) e^{.02s} + \frac{3/4}{.02}. \quad (29)$$

Validated endpoints give $x_1 \in [.5177014240300721, .517701424030129]$, so wealth does not exit. Preference is $u_s = 2 + \theta s + .05W_s$. Since $|\theta| \leq .2$, a preference exit implies

TABLE I
ORIGINAL-ECONOMY SLICE: INSTANTIATED BOUNDARY AND INTERIOR CERTIFICATES

k	Final θ	L	m	Class regret upper	Total seconds
1/2	1/5	1.615	0.9622	0	38.94
2	37/200	3.086	2.432	6.157×10^{-6}	39.5

The objective and stopping rule are those of the original economy; consumption, investment, restart, and constant-adjustment class are fixed as stated. m is a validated lower bound on $-F''$. Total time includes construction, derivative auditing, polls, and the final optimum check. It is not a full-domain policy certificate.

$\sup_{s \leq 1} |.05W_s| \geq .6$. Reflection and a Gaussian Chernoff bound give

$$p_{\text{exit}} \leq 4e^{-72} < 2.153 \times 10^{-31}. \quad (30)$$

This is an upper bound for every θ in the box, not an observed exit frequency.

The stopped objective is compared with an unstopped extension that clips preference to $[1.2, 2.8]$ inside utility and settlement. Their pathwise payoffs coincide off the exit event and have absolute value at most $H = 14.4$. A likelihood-ratio derivative on the full path, with score $W_1/.05$, bounds the payoff, first-derivative, and second-derivative differences. It avoids differentiation of a random first-exit time. The bounds, including the explicit θ -dependence of the adjustment cost, are proved in Appendix C.

We then integrate Gaussian moments analytically. On $|Z| \leq 10$, the flow expansion about $u = 2$ has $|u - 2| \leq .7$. A degree-100 power series and a degree-20 discount expansion yield rigorous geometric and exponential remainders. Gaussian tail moments control the complement, including the derivative score factors. MPFR directed arithmetic encloses every coefficient and polynomial evaluation; saved binary64 endpoints are rounded outward. The script records separate flow, terminal, stopping, and arithmetic enclosures rather than reporting one unexplained tolerance.

Eighty intervals of width $1/200$ cover the complete θ box. They certify strict concavity. For $k = 1/2$, $-F'' \geq .9622197$ and $|F''| \leq 1.6146843$; for $k = 2$, $-F'' \geq 2.4326157$ and $|F''| \leq 3.0850803$. The oracle is executed at meshes $1/25, 1/50, 1/100, 1/200$, with $\sigma = 1/100$ and $\kappa = 1/100000$. Every required width is checked, including at the active face. The fixed expansion orders are claimed only at these meshes.

For $k = 1/2$, the derivative remains positive at $\theta = 1/5$. Strict concavity therefore certifies the upper-face global optimum within the class. Its class regret is zero, although its ordinary gradient is not zero. For $k = 2$, a validated sign bracket locates a unique interior optimum near .183. The final poll is $\theta = 37/200$, with class regret at most 6.157×10^{-6} from $F'(\theta)^2/(2m)$. Thus a higher adjustment cost changes the economic solution from a binding upper control to an interior action. The result concerns actual stopped welfare, not the differentiable proposal loss.

The instantiation resolves the boundary exclusion and demonstrates a stopped-gradient bridge on this class. It does not instantiate the financed 47-dimensional gradient bridge, and it does not show that fixed c and p are optimal among unrestricted controls. Those are distinct mathematical statements.

6. UNKNOWN-SOLUTION BELLMAN DISCOVERY AND COMPLETION

6.1. A coupled stopped inventory economy

There are d goods and eight decision periods, with discount $99/100$. Inventory states are $x \in \{0, \dots, L - 1\}^d$. A feasible order purchases zero units or one or two units of one good,

never exceeding capacity before demand. Demand is equiprobable over $d + 2$ vectors: zero, the all-ones vector, and each vector with one component equal to two and all other components equal to one. These common demand realizations couple the inventories.

For order a , let $z = x + a$, sales $s = \min(z, D)$, remaining stock $y = (z - D)_+$, and shortage $h = (D - z)_+$. The period reward is

$$r(x, a, D) = \sum_j s_j - \frac{3}{5} \sum_j a_j - \frac{1}{10} \mathbf{1}_{\{a \neq 0\}} - \frac{1}{50} \sum_j y_j^2 - \frac{2}{5} \sum_j h_j^2 - \frac{1}{20d} \left(\sum_j h_j \right)^2. \quad (31)$$

The last term makes aggregate shortage costly beyond the sum of itemwise shortages. The all-zero state stops immediately and pays $-2 - d/5$. Other terminal states pay $\sum_j x_j/5 - \sum_j x_j^2/50$. The stopping state cannot order its way back into the continuation economy. Hence both common shocks and the ability to avoid joint stockout matter for a policy's worst-state value.

The model is specified by rational primitives. It has no engineered value function or action oracle. We study $(d, L) = (2, 4), (3, 4), (4, 4)$, with an additional $(4, 5)$ resolution and capacity control. These yield 16, 64, 256, and 625 states. Changing L changes the stated finite economic model; it is not an estimate of a continuous-state discretization rate.

6.2. Equal information and reference ordering

Raw policies are obtained by backward fitted Bellman recursion. At each stage a candidate constructs action values using its own preceding value approximation and takes the maximum over feasible actions. The targets are not taken from an optimal reference. Neural candidates use tanh networks of width 16 or 32 and one or two hidden layers, with seeds 27301 and 27302. Each fitted stage uses 400 Adam steps at .01, followed by at most 150 L-BFGS-B objective-gradient calls. The last accepted iterate is restored if the call cap is reached. Polynomial candidates use total degrees two and three and least squares on exactly the same states and candidate-specific Bellman targets. All candidates have the same primitives, transition information, terminal payoff, and stopping convention.

There are eight neural configurations and two polynomial configurations at each $L = 4$ model. The $d = 4, L = 5$ control uses the two width-32, depth-two neural seeds and both polynomial degrees, giving 34 retained cases in total. The period-zero action is compiled but its value approximation is not fitted, since no predecessor would use it. This explicit implementation reduction gives seven fitted stages rather than eight and changes no period-zero action target.

For each case, the complete raw action table is saved and hashed before certification. Exact own-policy evaluation produces (13) and (14). Completion uses a freshly constructed model and only the frozen action table, with the same $\varepsilon = 1/100$ for every case. It saves the changed actions, their original suffix-conditioned deficits, the completed policy, and a second reference-free certificate. All candidates and both certificates for a model are frozen before a separately coded scalar-integer optimal recursion is invoked. That recursion does not call the vectorized action-value or maximization routines used by the certificate. It verifies the policy gaps afterward; it supplies no labels, features, stopping decisions, or hyperparameters to training.

This is a designed experiment, not a preregistered clinical trial. The base R27 model was frozen before the present executions, but its completion amendment was explicitly written after prior diagnostic work. R28 inherits that post-diagnostic algorithmic choice and applies it to every new case without selection. The present source commit, run environment, all weights, all policies, and their freeze-order hashes are recorded. No claim of confirmatory independence from the earlier model-design process is made.

TABLE II
UNKNOWN-SOLUTION INVENTORY: ALL CASES, ALL STATES, ALL RESTARTS

d	L	States	Cases	Raw passes	Completed passes	Worst completed bound
2	4	16	10	8	10	2.089×10^{-4}
3	4	64	10	4	10	0
4	4	256	10	0	10	0
4	5	625	4	0	4	3.962×10^{-4}

A pass means the exact-rational reference-free bound is at most .01. Counts are computational coverage, not independent statistical replications. The complete 34-case table, raw bounds, exact gaps, changed-action counts, tolerance profile, architectures, and timings appear in the supplement.

TABLE III
INVENTORY COMPUTATION: MEASURED WORK AND DIRECT-REFERENCE COST

d, L	Neural train	Poly train	Raw check	Completion	Completed check	Exact DP
2,4	1.687	6.819×10^{-4}	0.001872	0.001072	0.00179	0.001284
3,4	1.778	9.284×10^{-4}	0.007657	0.004351	0.007219	0.007796
4,4	2.06	0.002249	0.03527	0.01918	0.03299	0.04531
4,5	4.134	0.003528	0.08468	0.04699	0.0806	0.1218

Seconds; candidate-stage columns are within-model medians, with neural and polynomial generation separated. Exact DP is the independently executed scalar optimal recursion and excludes candidate training. Construction, compilation, independent gap evaluation, total wall time, model-array bytes, action-table bytes, and process high-water memory are reported separately in the supplement. Shared reference costs are shown both unamortized and per case; no reference cost is silently assigned zero.

6.3. All-state accuracy, correction burden, and classical work

All 34 completed candidates satisfy the uniform target. The largest completed envelope is 3.962×10^{-4} , while 12 raw candidates already pass. We retain the raw failures rather than treating completion as evidence that neural training alone attains the tolerance. The independent optimal recursion confirms every saved envelope at every state and restart. The gap between mean fitting accuracy and a worst-restart guarantee becomes visible as the state model grows: local action mistakes at financially fragile inventory states can propagate backward, and completion corrects those mistakes against the completed suffix rather than against an unavailable optimal continuation.

Two distinctions govern the interpretation. First, completed policies can contain nonneural corrections. The supplement reports the number and fraction of nonstopped decisions changed, so the burden of correction is observable. Second, exact tabular backward induction is extremely competitive at these sizes. It is shown as a standalone reference solve in the cost table, not hidden inside the certificate’s timing. The experiment establishes unknown-solution discovery plus certified completion; it does not establish that neural approximation beats tabulation, polynomials, or adaptive sparse grids in a high-dimensional limit.

7. MECHANISM AND PRODUCTION-SCHEDULE EXPERIMENTS

7.1. Recentered historical transport

The new mechanism study uses seeds 27101–27103, proposal quadrature orders (8, 16) and (12, 24), 200 steps, and learning rate .08. One implementation uses the neural optimizer directly;

TABLE IV
INDEPENDENT IMPLEMENTATIONS OF THE EXACT RECENTERED IDENTITY

Seed	Proposal orders	Parameter difference	Output difference	Retained remainder
27101	(8,16)	2.466×10^{-12}	2.463×10^{-12}	1.451
27101	(12,24)	2.107×10^{-12}	2.123×10^{-12}	1.451
27102	(8,16)	5.045×10^{-13}	4.992×10^{-13}	1.392
27102	(12,24)	5.165×10^{-13}	5.125×10^{-13}	1.392
27103	(8,16)	1.072×10^{-12}	1.066×10^{-12}	1.503
27103	(12,24)	8.398×10^{-13}	8.648×10^{-13}	1.503

Maxima are over 200 steps. Carrier/economic mismatch is zero by construction and capped-step count is zero. The remainder is measured in quotient output norm. All first- and second-moment discrepancies and effective step distributions are retained. Representative policies from seed 27101 at both quadrature orders are separately checked by the stopped MPFR verifier.

the other independently computes the quotient gradient, Jacobian pullback, first and second moments, and the full parameter displacement. Economic coordinates are recentered on realized carrier outputs after every step. There is no output cap, and no nonlinear remainder is dropped.

The maximum parameter discrepancy across the six paths is 2.466×10^{-12} . This numerical agreement supports consistency of the implementations with Proposition 3. It does not certify equality of floating-point executions, and it does not identify a representation-free method: the carrier, its Jacobian, and all historical parameter moments remain present. In the earlier capped control, high nominal rates could produce the same effective trajectory because every step hit the cap. The new comparison removes that tuning confound rather than reinterpreting the old nominal-rate grid as uncapped rate identification.

7.2. *Production schedules, rejected proposals, and matched controls*

The production study uses seeds 27201 and 27202 and the fixed (8, 16) proposal rule. Neural Adam uses the inherited selected rate .08; direct quotient Adam uses .32. Direct L-BFGS-B provides a classical comparator. Each method has four blocks of at most 100 objective-gradient calls, with otherwise identical gated and ungated arms. Adam retains its moments across accepted blocks. L-BFGS-B is restarted in both arms at each block, so restart treatment is not a confound. Both arms pay for the same initial and block-end verifications; the ungated arm simply deploys the candidate regardless of the interval comparison.

The gate uses strict payoff separation with no injected extreme learning rate. A rejected block restores the network or quotient parameters, all Adam moments and counters where applicable, and all saved optimizer settings. Saved, candidate, restored, and deployed state hashes are retained, together with the candidate’s actual payoff interval, incumbent interval, strict margin, and interval width. These are causal online trajectories, not replayed selections from a completed path.

There are 6 actual rejections in the gated trajectories, all with exact restoration of the saved optimizer state. The direct L-BFGS-B paths exercise late-stage rejection under a production schedule; the Adam paths need not reject in order for the safety proposition to hold. Importantly, interval overlap can cause rejection of an economically indistinguishable or slightly better proposal. We therefore do not call these counts a measured frequency of harmful proposals. The paired intervals in the supplement determine which improvements, if any, are strictly certified between final gated and ungated policies. Absent strict separation, a lower printed regret upper bound is not proof that rollback improves payoff.

TABLE V
FIXED-SCHEDULE CENTRAL-STATE DEPLOYMENT: ALL TWELVE TRAJECTORIES

Seed	Method/arm	Calls	Rejections	Final regret upper	Total seconds
27201	Neural Adam / U	400	0	0.001926	88.54
27201	Neural Adam / G	400	0	0.001926	88.33
27201	Direct Adam / U	400	0	0.001937	88.53
27201	Direct Adam / G	400	0	0.001937	88.68
27201	Direct L-BFGS-B / U	114	0	0.001924	87.69
27201	Direct L-BFGS-B / G	136	3	0.001924	87.71
27202	Neural Adam / U	400	0	0.001926	88.52
27202	Neural Adam / G	400	0	0.001926	88.82
27202	Direct Adam / U	400	0	0.001937	88.44
27202	Direct Adam / G	400	0	0.001937	88.31
27202	Direct L-BFGS-B / U	107	0	0.001924	87.82
27202	Direct L-BFGS-B / G	115	3	0.001924	87.64

The payoff is at the central restart in the financed class. Total seconds are measured same-process, warm-infrastructure wall times, including initial and checkpoint verification and file bookkeeping. They are not cold from-scratch times; historical dual construction is separately disclosed. A rejected interval comparison need not mean the candidate truly loses payoff.

The results do not displace direct L-BFGS-B from the strongest tested central-state frontier. They establish that the same certificate gate is implementable for neural and direct methods, and that full-state rollback can be exercised without an adversarial rate injection. An algorithmic efficiency claim would require the gains from avoided losses to exceed verification and restoration costs over a broader population of economically relevant problems. That claim is not inferred from an implementation test.

8. CONTINUOUS VERIFICATION, ORIGINAL-DOMAIN PROGRESS, AND WORK ACCOUNTING

8.1. Retained stochastic verification benchmark

The inherited three-dimensional state–time benchmark has two independent Brownian sources, stopping, and an unspanned state. It is intentionally manufactured: its value witness and optimal actions are known and used in training. We retain it to test whole-cell verification, not as evidence of unknown-solution discovery. Its details and exact Hamiltonian-square identity are reproduced in the supplement; the full earlier derivations remain in the historical annex.

For the two fixed neural policies, the R26 second-order interval verifier gives bounds .0008343368 and .0007884142 on 32,768 cells. The same-cell first-order bounds are .00137310 and .00133294. Thus the second-order enclosure genuinely sharpens verification for unchanged policies. The exact rational degree-eight comparator gives $1.2978908373252045 \times 10^{-7}$ on 64 one-dimensional cells. That comparator exploits known optimal structure and must not be confused with the information-matched polynomial discovery candidates in Section 6.

Neither this three-dimensional success nor the finite-state dimension study proves scalable continuous-state interval-Hessian verification. Mixed derivatives grow quadratically in dimension, rectangular covers can grow exponentially, and interval dependency can dominate. Our new finite-state study quantifies actual enumeration, architecture, tolerance, and storage at 16–625 states; it is not relabeled as a high-dimensional continuous verifier result.

8.2. What changed in the original full-domain bound

The original history is decomposed as follows:

Object	Regret upper	Identified change
Coarse retained object	7.2414624431	Original retained actor and witness, coarse verification.
Fine unchanged object	7.2004087464	Same actor and witness; finer verification only.
Fine revised object	7.1818345808	Changed actor/witness candidate, common finer verification.

The first decrease, about .041054, is verifier tightening. The second, about .018574 or .258% of the unchanged fine bound, is a joint actor/witness certificate improvement. It is not a certified increase in policy payoff. The historical artifact explicitly records that no such payoff comparison was proved. Moreover, the first continuation block used 82 gradient calls and 48 accepted optimizer iterations, whereas the next three blocks used one call each, accepted no optimizer iteration, and left the candidate hashes unchanged. This stagnation is an empirical observation, not the conclusion of Theorem 2 for the full-domain class.

The new original-economy slice establishes a different, correctly restricted guarantee. We do not splice its small regret into the full-state ledger or replace the target in (7). The full-domain bound and its unresolved payoff-improvement comparison remain visible in both the article and the referee disposition table.

8.3. Complete cost categories and the measured boundary

For a deployed method, the relevant cost is

$$C_{\text{total}} = C_{\text{construction}} + C_{\text{tuning}} + C_{\text{proposal}} + C_{\text{geometry}} + C_{\text{verification}} + C_{\text{completion}} \quad (32) \\ + C_{\text{rollback}} + C_{\text{reference}} + C_{\text{bookkeeping}}.$$

The inventory experiment separately measures model construction, fitted-stage generation, action compilation, own-policy certification, fresh-model completion, completed-policy certification, independent reference construction and solution, and final reference evaluation. The slice includes the power-series oracle construction and whole-box derivative audit. Neither uses a hidden optimal-value witness in candidate generation.

The central-state experiment intentionally reuses the frozen historical dual library and process-level validated quadrature cache. Its run time therefore excludes cold reconstruction of those inputs. The library’s original construction record reports 415.788436934 seconds for 20 freshly generated nodes, including policy and dual generation and verification, with maximum resident memory 509,348 KiB on its original machine. We retain that record and its input hashes, identify warm reuse explicitly, and do not add seconds from different machines into a purported new measured total. A deployment amortizing this library over K solves must additionally charge its reconstruction cost divided by K ; $K = 1$ is not free. The new scripts also record the fixed rates as inherited tuning choices rather than claiming zero-cost tuning from scratch.

Peak process memory is a high-water measure that includes the numerical runtime; it is not the sum of certificate-array bytes and model weights. Both object sizes and process memory are reported. No cell count, tuning run, or checkpoint is counted as an independent replication of a statistical neural effect. The complete ledger permits method comparisons at a common declared information and infrastructure boundary and makes the remaining cold-pipeline measurement explicit.

9. CANDIDATE ATTRIBUTION, ROBUSTNESS, AND STATE-SPACE SCALING

The new cohort holds the 34 R28 raw policies fixed and adds 81 candidates under a committed protocol. On each original model we add ten random feasible tables, a one-period greedy policy, a base-stock rule, and two-period truncated dynamic programming. The width-32, depth-two seed panel is expanded to ten seeds at $(d, L) = (4, 4)$ and five at $(4, 5)$. At dimensions five, six, and seven we run six fixed proposal rules each. No failed candidate is replaced. The seven models contain between 16 and 16,384 states per time slice; the largest is 64 times the original 256-state four-dimensional graph. This is a finite-graph scaling study, not a claim to solve a high-dimensional continuous problem.

Every new raw action table and training record is hashed before its own completion. All reference-free variants are frozen before the independent full-horizon optimal reference is constructed for that model. A runtime audit rejects optimal-reference file reads, and the candidate/completion context disables the independent optimal solver. The exact reference is generated by a separate scalar implementation. The program checks every state and restart against it, not only a start-state average. The process had access to prior R27/R28 outcomes, so these additions and the residual-budget rule are exploratory, despite their fixed cohort and enforced program dependencies. The missing original R27 raw-run artifacts have not been reconstructed; R28 and R29 are separately identified executions.

Table VI distinguishes true raw regret, a reference-free raw residual bound, and the guarantees after repair. There are 115 raw policies and 286 completed variants, including the 14-policy tolerance panel at .001, .01, and .05. All completed variants satisfy their exact rational tolerance. A failed raw residual test is not, by itself, proof that raw regret exceeds the tolerance; the independent reference determines the latter count. The budget rule makes fewer changes than the fixed rule in 60 main-cohort cases, more in 0, and the same number in 55. These are paired deterministic comparisons, not independent statistical replications.

TABLE VI
EXACT ALL-STATE/ALL-RESTART COVERAGE AT TOLERANCE .01

d/L	N	Cases	Raw bound	Raw regret	Fixed	Budget
2/4	16	23	8	8	23	23
3/4	64	23	4	4	23	23
4/4	256	31	0	0	31	31
4/5	625	20	0	0	20	20
5/4	1024	6	0	0	6	6
6/4	4096	6	0	0	6	6
7/4	16384	6	0	0	6	6

The control experiment makes attribution observable. Random, myopic, base-stock, and two-period policies receive exactly the same two completion operators as neural and polynomial proposals. Detailed corrections by time, inventory total, minimum inventory, and number of empty coordinates are retained, together with suffix-conditioned deficit quantiles. Thus a small correction fraction can be distinguished from corrections concentrated near default or at early restart times. Raw-to-completed payoff monotonicity follows from Theorem 3, not from a reduction in a certificate alone.

Completion cost need not be proportional to the number of corrected actions: even a policy requiring no changes incurs the model-wide action scan. A small neural correction fraction therefore does not establish a computational advantage. Table VII retains exact dynamic programming

visibly and charges both repair and verification. All times within that table are measured on the same R29 execution environment. A dash means that no new neural fitting run was specified for that model; inherited R28 fitting times are not combined with R29 hardware timings. Full candidate-specific ledgers are in the computational appendix. In the largest graph, the specified neural proposal requires 634 replacements out of 131064 nonsettled state–time decisions under the budget rule, compared with 114341 for the myopic control. Its raw worst regret is 1.549, compared with 3.659 for that control. This is a substantial candidate-preservation difference in the executed model. It is not a total-work advantage: the neural fitting stage costs 51.6 seconds, whereas the independent full optimal solve costs 5.495 seconds in the same execution.

TABLE VII
SAME-EXECUTION STAGE COSTS (SECONDS; MEDIAN OVER THE SPECIFIED POLICIES)

d/L	N	Exact DP	New neural	Fixed repair	Budget repair	Verify
2/4	16	0.001072	–	0.001992	0.002268	0.001611
3/4	64	0.00803	–	0.007419	0.008217	0.006628
4/4	256	0.03729	2.291	0.02754	0.03296	0.03244
4/5	625	0.103	3.28	0.08989	0.1022	0.08187
5/4	1024	0.2064	4.424	0.1365	0.1498	0.1556
6/4	4096	1.103	13.33	0.6438	0.6971	0.7865
7/4	16384	5.495	51.6	3.282	3.677	4.668

The logical work count is $T \sum_s |A(s)|D$ successor contributions for one feasible-action sweep. The vector implementation also records the larger dense count $TNAD$, model-array bytes, compiled-policy bytes, integer bit lengths, CPU time, wall time, and process high-water resident memory. High-water memory includes libraries, previous cases, and live models; it is not an isolated per-candidate peak. The standalone component sum charges the full model and independent reference to each policy without amortization. Interpreter startup and file serialization are included in the outer cohort measurement but not silently allocated into individual method timings. Reusing an inherited candidate does not make its historical training free; those rows are explicitly marked as replay-only rather than new standalone fitting measurements.

The original economy remains distinct. The two rigorously enclosed preference-adjustment slices, six independently implemented transport paths, and twelve paired production trajectories are reproduced from the frozen R28 evidence. They instantiate boundary-aware polling, the stopped likelihood-ratio bridge for the stated scalar class, and full-state rollback. They do not establish the financed 47-dimensional derivative bridge or change the original current-state, whole-domain bound 7.181834580823298 against .01. Nor does the present cost ledger recover the unmeasured construction time of inherited central dual infrastructure. These are identified mathematical and measurement requirements, not replaced by auxiliary success.

10. CONCLUSION

A Bellman method must identify its approximation object, control class, restart set, verification oracle, and total computational work. The residual-budget theorem constructs a uniformly accurate finite-state policy while preserving every candidate action that its suffix envelope can certify. The constrained-poll theorem and stopped slice give a separate continuous-control guarantee, including boundary stationarity and oracle-precision costs. The exact transport identity and paired deployment experiment identify mechanisms without attributing generic repair to neural fitting. These contributions retain the original economic problem and its .01 target.

They do not turn a finite graph into a continuous-domain certificate, and they do not declare the original 7.181834580823298 bound closed. The complete computational and historical appendices retain all unsuccessful candidates, detailed derivations, and prior evidence.

APPENDIX A: PROOFS OF THE CERTIFICATE AND COMPLETION RESULTS

PROOF OF PROPOSITION 1: Apply stopped Itô’s formula to $e^{-.04(s-t)}v(s, u_s, x_s)$, first on compactly localized cylinders and then pass to the stopping time using the stated boundedness. For any admissible a ,

$$J^a - v(t, u, x) = \mathbb{E} \left[e^{-.04(\tau-t)}(g - v)(\tau, u_\tau, x_\tau) + \int_t^\tau e^{-.04(s-t)} \mathcal{R}^a v(s, u_s, x_s) ds \right]. \quad (33)$$

Since $\mathcal{R}^a v \leq e + q$, $J^a \leq v + b + (e + q) \frac{1 - e^{-.041-t}}{.04}$. For the candidate, $J^\pi \geq v - b - e \frac{1 - e^{-.041-t}}{.04}$. Take the supremum in the first inequality and subtract the second. Admissibility gives the nonnegative lower bound on regret. This argument uses a smooth test witness, not a differentiable optimal value. *Q.E.D.*

PROOF OF THEOREM 1: The terminal gap is zero. Suppose $0 \leq V_{t+1}^* - v_{t+1}^\pi \leq B_{t+1}^\pi$. Monotonicity of the Bellman operator gives the lower bound and

$$V_t^* - v_t^\pi = T_t V_{t+1}^* - T_t v_{t+1}^\pi + \delta_t^\pi \quad (34)$$

$$\leq \beta \max_a P_t^a B_{t+1}^\pi + \delta_t^\pi = B_t^\pi. \quad (35)$$

The inequality follows by comparing each action’s continuation difference before taking the maximum. At a stopped state there is no continuation, so the gap and the envelope are zero.

For completion, proceed backward after fixing the completed suffix. If the raw action is retained, its deficit against that suffix is at most η by the test. If it is replaced by a maximizing action, the deficit is zero. Thus $\delta_t^\pi \leq \eta$ everywhere. Induction in the previous bound yields $B_t^\pi \leq \eta \sum_{j=0}^{T-t-1} \beta^j$, proving the assertion. This argument does not assume that the raw policy is neural, near optimal, or generated independently of the model. Reference-free means that the implemented construction never reads V^* ; the proof may of course compare to the mathematically defined optimal value.

Each backward sweep enumerates at most TNA action values, each with at most D successors. A maximization and a retained-action test add linear overhead. When primitives are rational, a common denominator can be propagated backward and all comparisons reduced to integer comparisons. Their bit cost is not a unit-cost real operation and must be counted separately for growing horizons or large denominators. *Q.E.D.*

A.1. Exact arithmetic convention in the inventory implementation

Let $R = 100d$, $D = d + 2$, and let r_{saz}^{num} be the integer reward numerator. Terminal values have denominator $q_T = R$. Define $q_t = 100Dq_{t+1}$. If the next value numerator is n_{t+1} , the action numerator is

$$Q_{t,s,a}^{\text{num}} = 100 \frac{q_{t+1}}{R} \sum_{z=1}^D r_{saz}^{\text{num}} + 99 \sum_{z=1}^D n_{t+1}(s'(s, a, z)). \quad (36)$$

The stopped action is instead assigned its settlement numerator at denominator q_t . Equation (36) exactly equals the rational expectation and discount recursion. Python arbitrary-size integers

carry all accumulated numerators. Binary64 training is allowed to choose a poor action; it is not allowed to change a certificate comparison. Exported decimal bounds are rounded upward for display, whereas the decision uses the exact rational fraction. The independent optimal recursion uses a scalar outcome loop and separately selects feasible maximizers. Every saved policy gap is then compared with its saved statewise envelope using common integer denominators.

PROOF OF PROPOSITION 2: Containment gives $J(\pi_{\text{candidate}}) \geq L(\pi_{\text{candidate}}) > U(\pi_{\text{incumbent}}) + \zeta \geq J(\pi_{\text{incumbent}}) + \zeta$. A rejected candidate is not deployed. Restoring a byte-equivalent serialization of every deterministic optimizer state component makes the next state identical to the saved state; restoring only policy parameters would not establish this fact for an adaptive optimizer. Summing the strict improvements and using $J \leq M$ gives the finite-acceptance bound when $\zeta > 0$. *Q.E.D.*

APPENDIX B: CONSTRAINED POLLING: GEOMETRY, PRECISION, AND WORK

PROOF OF THEOREM 2: Write $w = \kappa h^2$. Failure of the strict acceptance test at a feasible trial y means $L(y) \leq U(x) + \sigma h^2$. Since intervals contain the true values and each has width at most w ,

$$F(y) - F(x) \leq (\sigma + 2\kappa)h^2. \quad (37)$$

For $y = x + she_i \in C$, $s \in \{-1, 1\}$, smoothness gives

$$F(y) - F(x) \geq sh \partial_i F(x) - \frac{L}{2}h^2. \quad (38)$$

Thus $s\partial_i F(x) \leq Ch$, where $C = \sigma + 2\kappa + L/2$.

For each i with nonzero derivative, take $s = \text{sgn } \partial_i F(x)$. If the full signed step is feasible, the preceding inequality bounds $|\partial_i F(x)|$, and scalar projection implies $|(G_\tau(x))_i| \leq Ch$. If it is infeasible, the distance to the face in that sign is less than h , and the same scalar projection gives $|(G_\tau(x))_i| < h/\tau$. A zero-width coordinate has projected component zero. Squaring and summing establishes (20).

Every accepted step improves true payoff by more than σh^2 , by the same containment argument as Proposition 2. Therefore the number of accepted steps is bounded by $\lfloor (M - F(x_0))/(\sigma h^2) \rfloor$. Each accepted or final unsuccessful poll requires no more than one incumbent evaluation and $2n$ trial evaluations. Multiplying by the stated per-evaluation work gives (19). Computational failure at a feasible trial destroys the failed-poll inequality for that direction; it cannot be classified as a geometric short face. This proves the final qualification as well. *Q.E.D.*

For the constant-control slice, $M = 0$ is a valid upper payoff bound: flow utility and adjustment costs are nonpositive, while terminal wealth is below one and any earlier boundary settlement includes a nonpositive early-exit penalty. A lower endpoint for the initial payoff supplies a conservative $M - F(x_0)$. At the four executed meshes the code checks every required width, and a payoff call evaluates a degree-100 interval polynomial after construction. Its arithmetic cost is linear in the degree; setup, coefficient construction, and derivative auditing are separate. Measured wall time is an observation, not a proved worst-case CPU-time oracle. The large sufficient-decrease work bound is not presented as a sharp predictor of the observed call count.

For completeness, the sign of the settlement payoff in this upper-bound argument can be checked without a numerical approximation. Wealth in the slice is

$$x_s = \frac{75}{2} - \frac{145}{4}e^{s/50}, \quad x_{1/2} < \frac{75}{2} - \frac{145}{4} \left(1 + \frac{1}{100}\right) = \frac{71}{80} < 1.$$

The strict inequality uses $e^z > 1 + z$ for $z > 0$. Wealth is decreasing. If $s \leq 1/2$, settlement is at most $.1 \log(5/4) - 4 < 0$. If $s \geq 1/2$, wealth is below one, the quadratic preference term is nonpositive, and the early-exit charge is nonpositive; settlement is again nonpositive. The same conclusion holds at the terminal time. Running utility and the adjustment-cost term are nonpositive throughout the stopped domain. Thus the complete stopped payoff is nonpositive pathwise, which justifies $M = 0$ in the finite-work bound independently of the computed payoff intervals.

The unconstrained positive-spanning result is recovered when all poll directions are feasible. At a binding upper face an increasing objective can have a nonzero ordinary gradient but a zero projected-gradient mapping. This is precisely the $k = 1/2$ case in the original slice. In general, neither projected stationarity nor a complete unsuccessful poll implies global optimality without further structure. The slice adds verified strong concavity; the finite-state completion result instead uses Bellman comparison and does not require parameter-space convexity.

APPENDIX C: THE STOPPED LIKELIHOOD-RATIO BRIDGE AND ANALYTIC ORACLE

C.1. Stopping and explicit parameter dependence

In the slice, x_s decreases monotonically and its interval endpoint remains above .5. Thus only preference can cause early exit. The drift moves preference by at most .2 over the horizon, leaving a Brownian excursion of at least .6 to reach either boundary from two. The reflection bound $\Pr(\sup_{s \leq 1} |W_s| \geq 12) \leq 4 \Pr(W_1 \geq 12)$ and $\Pr(W_1 \geq 12) \leq e^{-72}$ give (30).

Let H_θ be the stopped payoff as a functional of the preference path, and $\tilde{H}_\theta$ its full-horizon clipped extension. The path functionals agree when no exit occurs. Both are bounded in absolute value by $H = 14.4$ for the stated parameters: clipped flow utility is at most $5(4/3)^{1/5} < 6$ in magnitude, adjustment cost is at most .04 when $k \leq 2$, and the boundary/terminal settlement is bounded by the early-exit charge eight plus the bounded quadratic and logarithmic terms. At a fixed preference path, the only explicit θ dependence is the quadratic cost. The difference $D_\theta = H_\theta - \tilde{H}_\theta$ therefore satisfies

$$|D_\theta| \leq 2H \mathbf{1}_A, \quad |\partial_\theta D_\theta| \leq k(.2) \mathbf{1}_A, \quad |\partial_{\theta\theta} D_\theta| \leq k \mathbf{1}_A, \quad (39)$$

where A is the exit event for that path.

Changing the constant drift of the full preference path yields the likelihood-ratio score $S = W_1/\sigma$, $\sigma = .05$, under the current drift measure. The second score is $S^2 - 1/\sigma^2$. Bounded path functionals, their displayed explicit derivatives, and Gaussian exponential moments justify differentiating the expectation twice. Consequently,

$$|\mathbb{E} D_\theta| \leq 2Hp, \quad (40)$$

$$|\partial_\theta \mathbb{E} D_\theta| \leq 2H\sqrt{p}/\sigma + k(.2)p, \quad (41)$$

$$|\partial_{\theta\theta} \mathbb{E} D_\theta| \leq 2H\sqrt{2p}/\sigma^2 + 2k(.2)\sqrt{p}/\sigma + kp, \quad (42)$$

where $p = 4e^{-72}$. The square roots follow from Cauchy–Schwarz, $\mathbb{E} S^2 = 1/\sigma^2$, and $\mathbb{E}(S^2 - 1/\sigma^2)^2 = 2/\sigma^4$. The argument differentiates a change of path measure, not the hitting-time map. The explicit cost derivatives cannot be dropped from these bounds.

C.2. Power series and Gaussian moments

Set $a = \log(4/3)$ and $v = u - 2$. With $c = 3/4$, untruncated flow utility is

$$f(v) = -\frac{4}{3} \frac{e^{av}}{1+v} = -\frac{4}{3} \sum_{n=0}^{\infty} c_n v^n, \quad c_0 = 1, \quad c_n = \frac{a^n}{n!} - c_{n-1}. \quad (43)$$

The absolute coefficient of f is bounded by $(4/3)e^a = 16/9 < 2$. On $|Z| \leq 10$, $v = \theta s + \sigma\sqrt{s}Z$ satisfies $|v| \leq .7$, so a degree- N truncation has error at most

$$d_N = \frac{2(.7)^{N+1}}{1-.7}. \quad (44)$$

For $N = 100$ this is approximately 1.509423×10^{-15} . There is no singularity in this central region. Clipping handles the Gaussian tails without evaluating $1/(1+v)$ beyond its valid economic interval.

The full Gaussian polynomial moments are

$$\mathbb{E}(\theta s + \sigma\sqrt{s}Z)^n = \sum_{\substack{0 \leq j \leq n \\ j \text{ even}}} \binom{n}{j} \theta^{n-j} \sigma^j (j-1)!! s^{n-j/2}. \quad (45)$$

The convention $(-1)!! = 1$ is used. Each time moment $I_m = \int_0^1 e^{-\rho s} s^m ds$, $\rho = .04$, is enclosed by

$$I_m = \sum_{\ell=0}^K \frac{(-\rho)^\ell}{\ell!(m+\ell+1)} + r_{m,K}, \quad |r_{m,K}| \leq \frac{e^\rho \rho^{K+1}}{(K+1)!}, \quad K = 20. \quad (46)$$

The code propagates this remainder into every coefficient, including the running adjustment cost. The untruncated terminal expectation is the exact quadratic $-.02(\theta^2 + \sigma^2)$ plus $.1 \log x_1$, discounted at ρ .

C.3. Tail control for values and derivatives

Let T_j bound $\mathbb{E}[|Z|^j \mathbf{1}_{\{|Z|>10\}}]$. Dropping the normal density's factor $(2\pi)^{-1/2} < 1$ gives the rigorous recursions

$$T_0 = \frac{2e^{-50}}{10}, \quad T_1 = 2e^{-50}, \quad T_j = 2 \cdot 10^{j-1} e^{-50} + (j-1)T_{j-2} \quad (j \geq 2). \quad (47)$$

Define

$$P_r = 2 \sum_{n=0}^{100} \sum_{j=0}^n \binom{n}{j} (.2)^{n-j} \sigma^j T_{j+r}. \quad (48)$$

This controls the polynomial tail weighted by $|Z|^r$ uniformly over time and the drift box. The clipped utility has magnitude below six. For a normal expectation at time s , drift differentiation gives factors $\sqrt{s}Z/\sigma$ and $s(Z^2 - 1)/\sigma^2$. Hence the flow errors of orders zero, one, and two are bounded respectively by

$$E_0^{\text{flow}} = d_N + 6T_0 + P_0, \quad (49)$$

$$E_1^{\text{flow}} = (d_N + 6T_1 + P_1)/\sigma, \quad (50)$$

$$E_2^{\text{flow}} = \{2d_N + 6(T_2 + T_0) + P_2 + P_0\}/\sigma^2. \quad (51)$$

We used $\mathbb{E}|Z| \leq 1$ and $\mathbb{E}|Z^2 - 1| \leq 2$ on the central part and bounded the discount integral by one. These are remainder bounds for expectations and their derivatives, not pointwise derivative bounds for a clipped function.

For the terminal quadratic, the clipped and untruncated versions coincide in the central region. On the tail their difference is at most $.0144 + .0001Z^2$, because $|\theta| \leq .2$ and the clipping radius is $.8$. Thus the terminal errors are

$$E_0^{\text{term}} = .0144T_0 + .0001T_2, \quad (52)$$

$$E_1^{\text{term}} = (.0144T_1 + .0001T_3)/\sigma, \quad (53)$$

$$E_2^{\text{term}} = \{.0144(T_2 + T_0) + .0001(T_4 + T_2)\}/\sigma^2. \quad (54)$$

Adding the likelihood-ratio stopping bounds gives the total analytic errors. Directed interval arithmetic adds coefficient and evaluation rounding. The first two derivatives of the resulting polynomial are evaluated on each entire drift cell; their analytic error budgets are then added. This is the source of the validated L and strong-concavity constant m used in the main text.

C.4. From derivatives to class regret

If $F'' \leq -m < 0$ on the whole interval, then for any feasible y ,

$$F(y) - F(x) \leq F'(x)(y - x) - \frac{m}{2}(y - x)^2 \leq \frac{|F'(x)|^2}{2m}. \quad (55)$$

The code uses an upper enclosure for $|F'(x)|$ and a lower enclosure for m . At the upper face, a nonnegative lower bound on F' and strict concavity show that F is increasing on the entire box, giving exact class optimality. The analogous statement holds at the lower face. A sign-changing interval bracket for F' identifies the unique interior optimum. These arguments certify optimality within the fixed consumption/investment/constant-adjustment class; they do not compare that class to arbitrary adapted controls.

APPENDIX D: HISTORICAL TRANSPORT AND THE FINANCING-GRADIENT FORMULA

PROOF OF PROPOSITION 3: Suppose the two implementations agree through step n . They then evaluate the same raw carrier output and the same economic quotient. By the chain rule their parameter gradients are identical, $\nabla_\theta L = B_n^\top g_n$. Their first and second moments, bias corrections, denominators, and parameter displacement are consequently identical. Evaluating the full network at the new parameters gives identical raw outputs and includes the entire nonlinear displacement. Exact recentering supplies the induction hypothesis for the next step. A linearized output update would not supply it unless its Taylor remainder vanished. Expanding the first-moment recurrence gives the historical sum in (22); Taylor's integral remainder gives its stated bound when the Jacobian is Lipschitz. Q.E.D.

The inherited financing calculation is retained because it identifies what remains to be instantiated for the general stopped gradient. Let $P(y, a) = B$ be the exact financing equation and P_N its proposal approximation. Suppose $P_a, P_{N,a} \geq d_* > 0$ on the relevant root segment and

$|P - P_N| \leq e_P$. Then the root error is at most $\delta = e_P/d_*$. The reduced gradient of an objective H is

$$H_y - H_a \frac{P_y}{P_a}. \quad (56)$$

At the exact and approximate roots write $A = H_y$, $B_1 = H_a$, $C_1 = P_y$, and $D_1 = P_a$. Let E_A, E_B, E_C, E_D enclose both approximation errors and the root-shift effects over distance δ , and let M_B, M_C bound the absolute values of both exact and approximate B_1, C_1 . The reduced-gradient discrepancy is at most

$$E_A + \frac{E_B M_C + M_B E_C}{d_*} + \frac{M_B M_C E_D}{d_*^2}. \quad (57)$$

Indeed, add and subtract the mixed products in $B_1 C_1 / D_1 - B_{1,N} C_{1,N} / D_{1,N}$, and use $|D_1^{-1} - D_{1,N}^{-1}| \leq E_D / d_*^2$. All four error terms and both magnitude bounds are needed. Finite payoff secants do not supply these derivative enclosures. R28 instantiates a different exact stopped-gradient bridge on the one-dimensional slice, not the constants in (57) along the financed 47-dimensional optimizer path.

APPENDIX E: ALL-RESTART DISCOUNTED SUFFIX ENVELOPES

Let a nonnegative residual envelope equal a_i on time slab $[t_i, t_{i+1}]$, with $\Delta_i = t_{i+1} - t_i$. The discounted suffix at a slab boundary obeys

$$C_i = a_i \frac{1 - e^{-\rho \Delta_i}}{\rho} + e^{-\rho \Delta_i} C_{i+1}, \quad C_m = 0. \quad (58)$$

At a restart t in that slab, put $r = t_{i+1} - t$. The suffix is

$$H_i(r) = \frac{a_i}{\rho} + \left(C_{i+1} - \frac{a_i}{\rho} \right) e^{-\rho r}, \quad 0 \leq r \leq \Delta_i. \quad (59)$$

Its derivative has a constant sign unless it is identically zero. The maximum within the slab is therefore attained at an endpoint. Taking the maximum over the slab endpoints proves the all-restart envelope used by the postprocessor. When a separate uniform boundary mismatch is present, its contribution must be added with the coefficient justified by the underlying stopped verification inequality; it is not absorbed into an unexplained legacy output field. The postprocessed all-restart number is consequently a theorem-backed use of the same slab bounds, not an empirical maximum over sampled restart times.

REFERENCES

- Audet, C., and J. E. Dennis, Jr. (2006): “Mesh Adaptive Direct Search Algorithms for Constrained Optimization,” *SIAM Journal on Optimization*, 17(1), 188–217. doi:10.1137/040603371. [2]
- Brown, D. B., and J. E. Smith (2014): “Information Relaxations, Duality, and Convex Stochastic Dynamic Programs,” *Operations Research*, 62(6), 1394–1415. doi:10.1287/opre.2014.1322. [2, 10]
- Brown, D. B., J. E. Smith, and P. Sun (2010): “Information Relaxations and Duality in Stochastic Dynamic Programs,” *Operations Research*, 58(4, part 1), 785–801. doi:10.1287/opre.1090.0796. [2, 10]
- Brumm, J., and S. Scheidegger (2017): “Using Adaptive Sparse Grids to Solve High-Dimensional Dynamic Models,” *Econometrica*, 85(5), 1575–1612. [2, 10]
- Byrd, R. H., P. Lu, J. Nocedal, and C. Zhu (1995): “A Limited Memory Algorithm for Bound Constrained Optimization,” *SIAM Journal on Scientific Computing*, 16(5), 1190–1208. doi:10.1137/0916069. [3, 10]
- Hur , C., H. Pham, A. Bachouch, and N. Langren  (2021): “Deep Neural Networks Algorithms for Stochastic Control Problems on Finite Horizon: Convergence Analysis,” *SIAM Journal on Numerical Analysis*, 59(1), 525–557. doi:10.1137/20M1316640. [2, 10]
- Jacot, A., F. Gabriel, and C. Hongler (2018): “Neural Tangent Kernel: Convergence and Generalization in Neural Networks,” *Advances in Neural Information Processing Systems*, 31. arXiv:1806.07572. [3]
- Judd, K. L. (1998): *Numerical Methods in Economics*. Cambridge, MA: MIT Press. [2, 10]
- Kingma, D. P., and J. Ba (2015): “Adam: A Method for Stochastic Optimization,” *International Conference on Learning Representations*. arXiv:1412.6980. [3]
- Kolda, T. G., R. M. Lewis, and V. Torczon (2003): “Optimization by Direct Search: New Perspectives on Some Classical and Modern Methods,” *SIAM Review*, 45(3), 385–482. [10]
- Kolda, T. G., R. M. Lewis, and V. Torczon (2006): “Stationarity Results for Generating Set Search for Linearly Constrained Optimization,” *SIAM Journal on Optimization*, 17(4), 943–968. [10]
- Kushner, H. J., and P. G. Dupuis (2001): *Numerical Methods for Stochastic Control Problems in Continuous Time*, second edition. New York: Springer. [2]
- Munos, R., and C. Szepesv ri (2008): “Finite-Time Bounds for Fitted Value Iteration,” *Journal of Machine Learning Research*, 9(27), 815–857. [2, 10]
- Rump, S. M. (2010): “Verification Methods: Rigorous Results Using Floating-Point Arithmetic,” *Acta Numerica*, 19, 287–449. [3, 10]
- Sirignano, J., and K. Spiliopoulos (2018): “DGM: A Deep Learning Algorithm for Solving Partial Differential Equations,” *Journal of Computational Physics*, 375, 1339–1364. doi:10.1016/j.jcp.2018.08.029. [2]
- Vicente, L. N. (2010): “Worst Case Complexity of Direct Search,” preprint 10-17, Department of Mathematics, University of Coimbra; Optimization Online, June 2010. The cited sufficient-decrease result concerns smooth directional direct search and does not include the stopped interval-oracle instantiation developed here. [2]